

YOLOv8 Coal Classification — Comprehensive Report

Experiment: coal_yolov8_m_balanced2

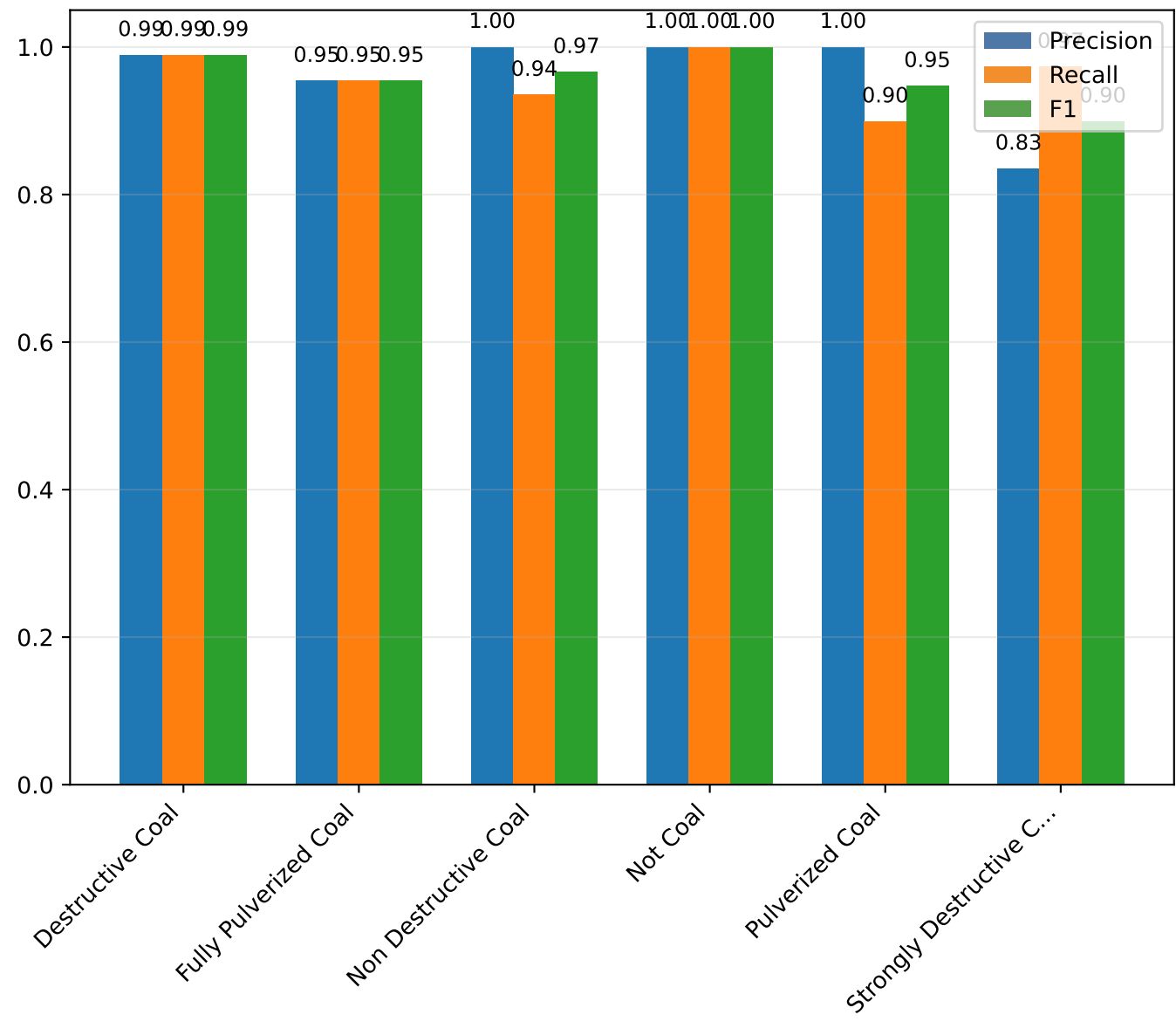
Generated: 2025-11-09 19:52:45

```
Overall_Accuracy: 0.9705400981996728  
Macro_F1: 0.959079509744425  
Weighted_F1: 0.9710911283445104  
mAP@0.5: 0.9904684669393088  
mAP@0.5:0.95: 0.9904684669393088  
FPS: 17.318722275103152  
Avg_Inference_ms: 57.740980201384055  
Accuracy_Speed_Balance: 0.7233118236464069  
Overall_Performance: 0.9800054812076808  
Production_Readiness: 0.5941639283056265
```

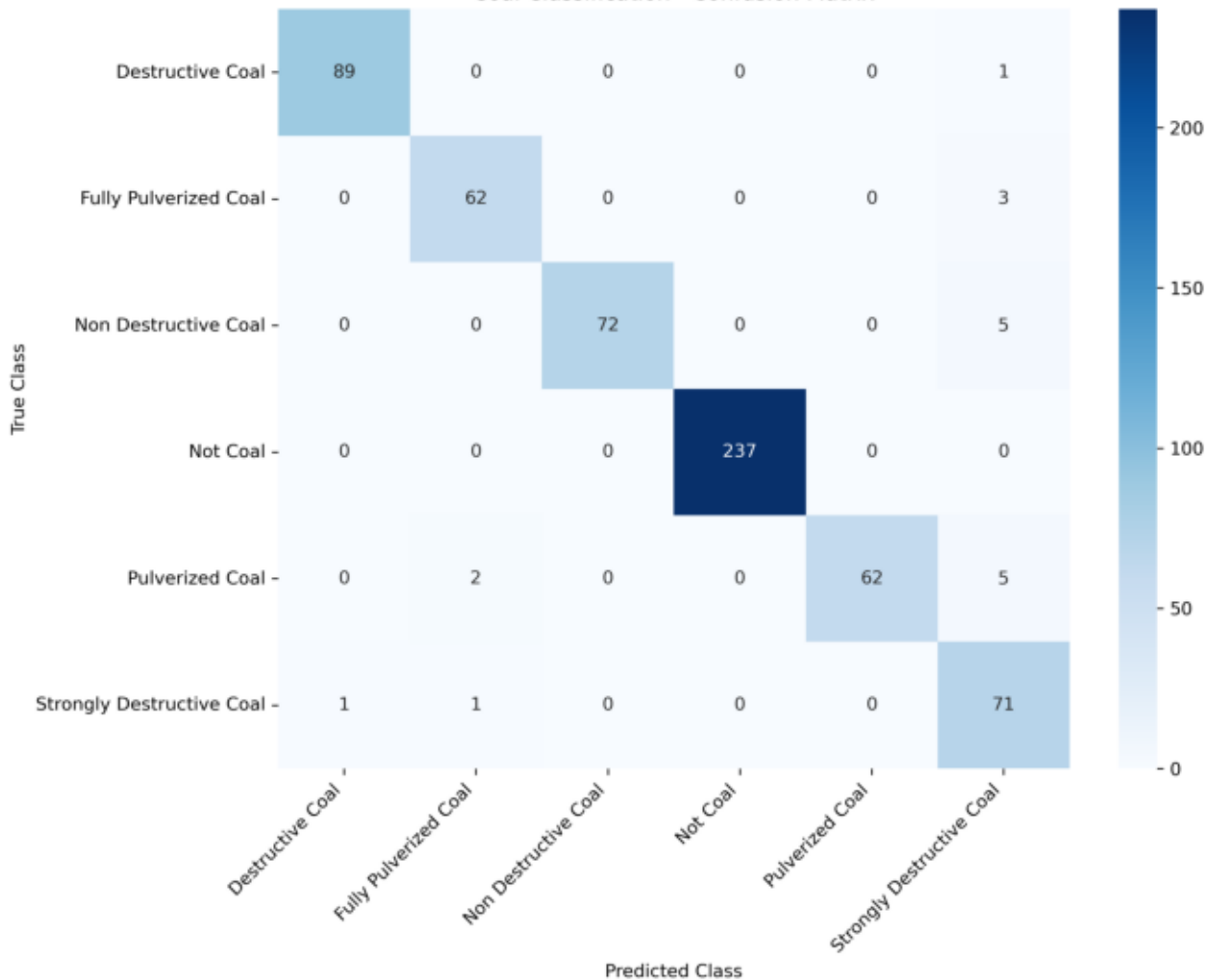
Experiment Configuration

agnostic_nms: False
amp: True
augment: False
auto_augment: randaugment
batch: 16
bgr: 0.0
box: 7.5
cache: False
cfg: None
classes: None
close_mosaic: 10
cls: 0.5
compile: False
conf: None
copy_paste: 0.5
copy_paste_mode: flip
cos_lr: True
cutmix: 0.0
data: E:\Yolo\YOLOv8_Coal_Classification_Project\data\train_balanced
degrees: 20
deterministic: True
device: 0
dfl: 1.5
dnn: False
dropout: 0.0
dynamic: False
embed: None
epochs: 20
erasing: 0.4
exist_ok: False
fliplr: 0.7
flipud: 0.0
format: torchscript
fraction: 1.0
freeze: None
half: False
hsv_h: 0.03
hsv_s: 0.8
hsv_v: 0.5
imgsz: 224
int8: False
iou: 0.7
keras: False
kobj: 1.0
line_width: None
lr0: 0.001
lrf: 0.01
mask_ratio: 4
max_det: 300
mixup: 0.4
mode: train
model: yolov8m-cls.pt
momentum: 0.937
mosaic: 0.7
multi_scale: False
name: coal_yolov8_m_balanced2
nbs: 64
nms: False
opset: None
optimize: False
optimizer: AdamW
overlap_mask: True
patience: 20
perspective: 0.0
plots: True
pose: 12.0
pretrained: True
profile: False
project: coal_classification_runs
rect: False
resume: False
retina_masks: False
save: True
save_conf: False
save_crop: False
save_dir: E:\Yolo\YOLOv8_Coal_Classification_Project\coal_classification_runs\coal_yolov8_m_balanced2
save_frames: False
save_json: False
save_period: 10
save_txt: False
scale: 0.8
seed: 0
shear: 0.0
show: False
show_boxes: True
show_conf: True
show_labels: True
simplify: True
single_cls: False
source: None
split: val
stream_buffer: False
task: classify
time: None
tracker: botsort.yaml
translate: 0.2
val: True
verbose: True
vid_stride: 1
visualize: False
warmup_bias_lr: 0.1
warmup_epochs: 3
warmup_momentum: 0.8
weight_decay: 0.0005
workers: 4
workspace: None

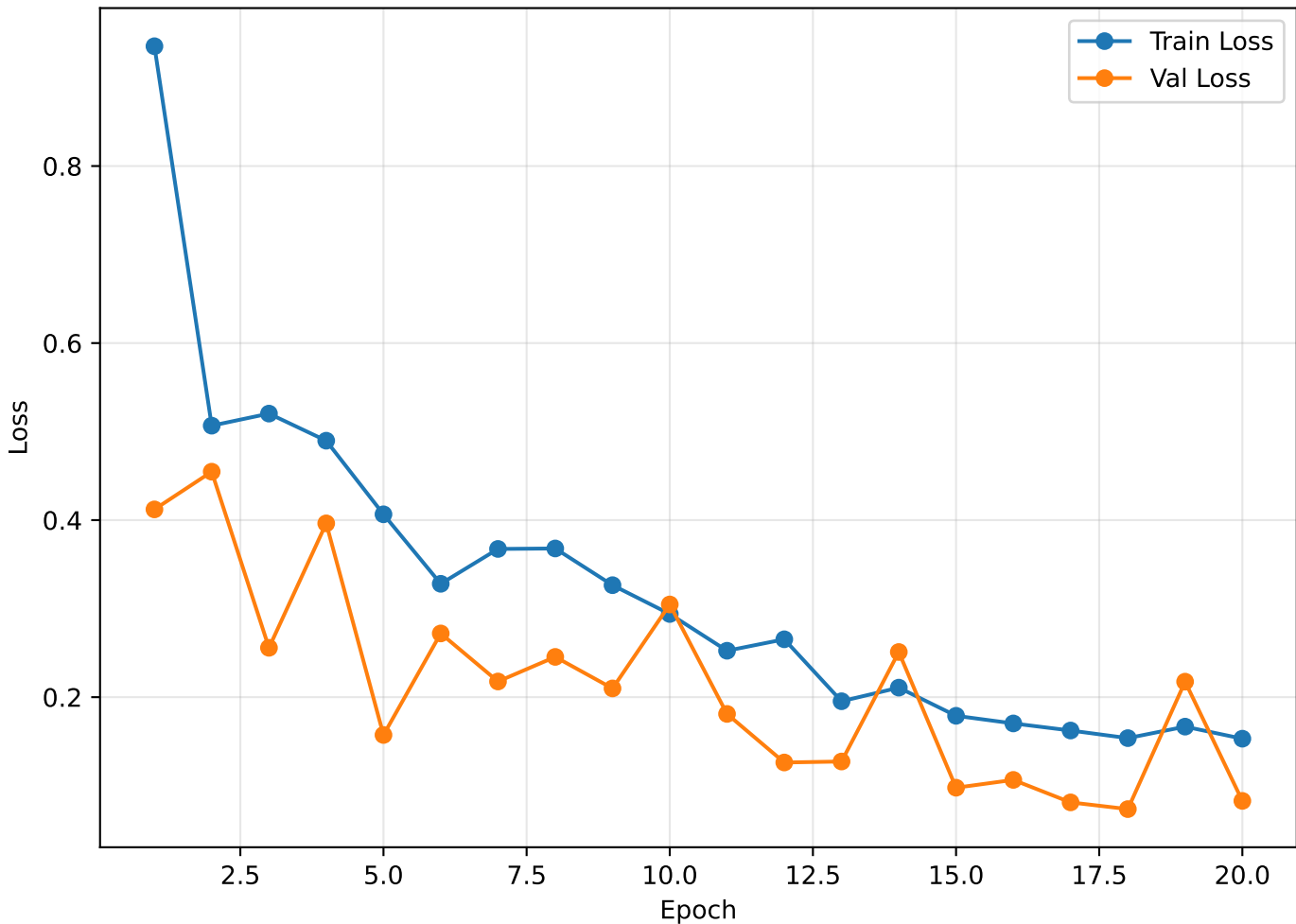
Per-class Metrics



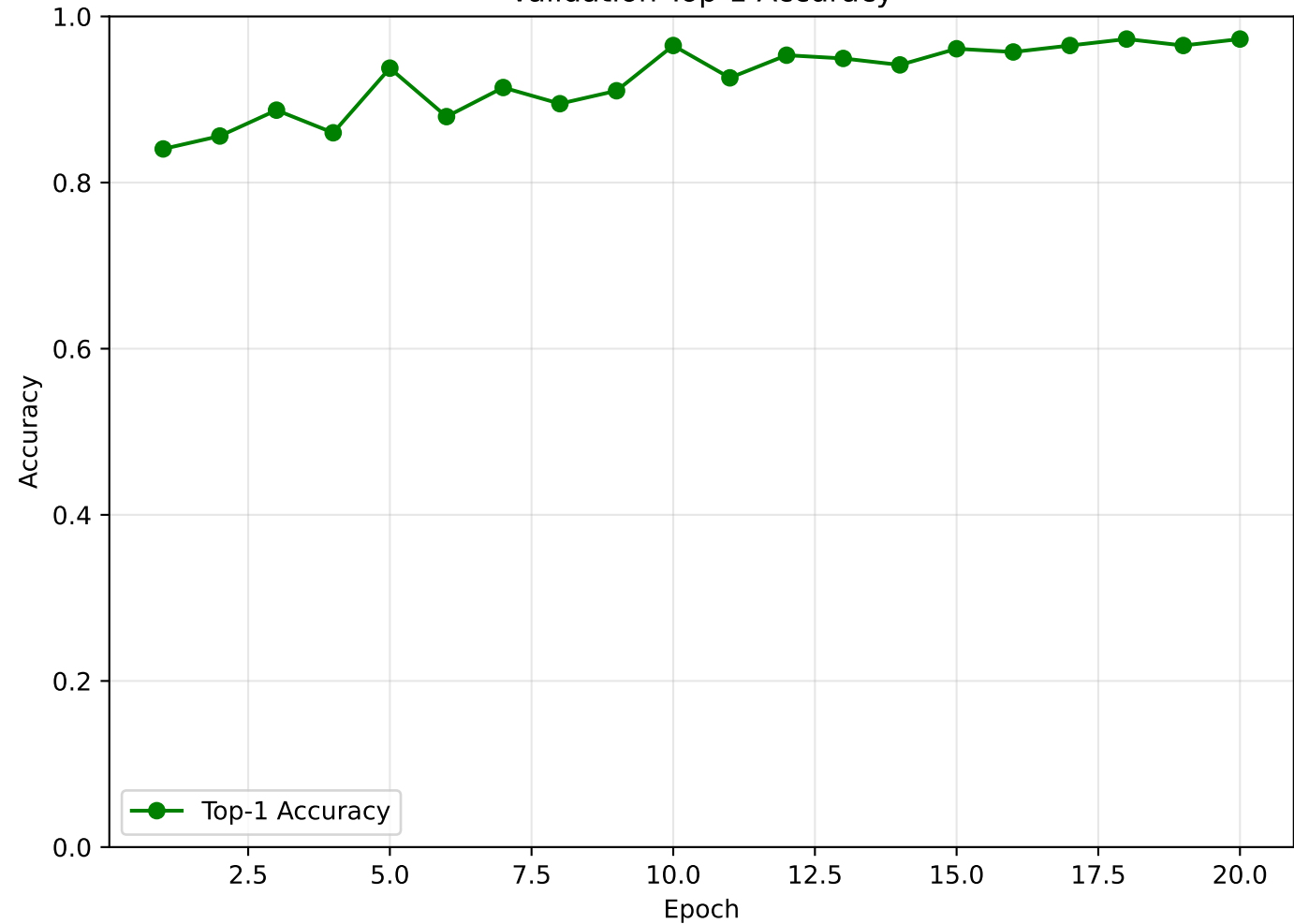
Coal Classification - Confusion Matrix



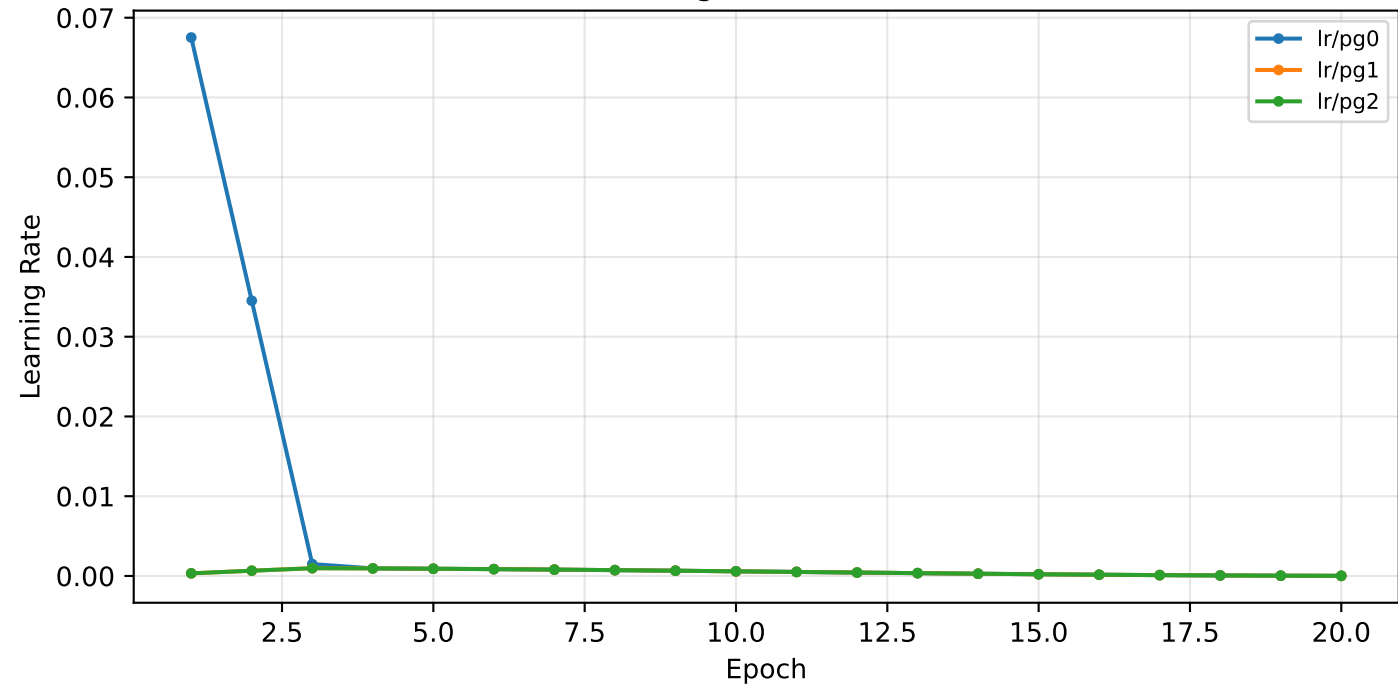
Training and Validation Loss



Validation Top-1 Accuracy



Learning Rate Schedule



Best training epoch summary:

epoch: 18.0

time: 280.089

train/loss: 0.15372

metrics/accuracy_top1: 0.97276

metrics/accuracy_top5: 1.0

val/loss: 0.07353

lr/pg0: 6.39518e-05

lr/pg1: 6.39518e-05

lr/pg2: 6.39518e-05

Detailed Per-Class Metrics

Class	Precision	Recall	F1_Score	Support	AP_Score
Destructive Coal	0.989	0.989	0.989	90	1.000
Fully Pulverized Coal	0.954	0.954	0.954	65	0.987
Non Destructive Coal	1.000	0.935	0.966	77	0.997
Not Coal	1.000	1.000	1.000	237	1.000
Pulverized Coal	1.000	0.899	0.947	69	0.984
Strongly Destructive Coal	0.835	0.973	0.899	73	0.975

Summary Metrics

Metric	Value
Overall_Accuracy	0.9705400981996728
Macro_F1	0.959079509744425
Weighted_F1	0.9710911283445104
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